

GLOBAL REGULARITY OF THE THREE-DIMENSIONAL NAVIER–STOKES EQUATION VIA THE BOREL SUBALGEBRA REDUCTION OF $E(4, 4)$

DREW REMMENG

ABSTRACT. The 4D incompressible Navier–Stokes (NS) equation has the exceptional Lie superalgebra $E(4, 4)$ as its complete symmetry algebra (Cantarini–Caselli–Kac, 2026). In the companion paper [?] we showed that the seed fiber of $E(4, 4)$ reduces the 4D problem to three surviving modes in the Verma module $M_1(1, 0, 0)$, and that the resulting dynamics are computationally universal (Σ_1 -hard) and A_∞ -wild [?].

Here we observe that the maximal parabolic subalgebra $\mathfrak{p} = \mathfrak{p}_{1,2,3} \subset \mathfrak{sl}_4$ (the stabilizer of the flag $V_3 = \text{span}\{e_1, e_2, e_3\}$) defines a dynamically invariant *3D slice*: initial data supported on wave vectors with $k_4 = 0$ evolve entirely within this slice, and the restricted dynamics are exactly the 3D incompressible NS equation on $\mathbb{T}^3$.

On this 3D slice the exceptional morphism ϕ_{4H} of degree 4 is *absent*: the symmetry algebra of the 3D equation has Complex B morphisms only in degrees 1 and 2. The Ext algebra $\text{Ext}^*(M_*(1, 0, 0)|_{3D}, M_*(1, 0, 0)|_{3D})$ is therefore a path algebra of a quiver of finite or tame type — the A_∞ obstruction of Paper 2 does not arise.

We prove:

- (a) **(Borel Slice)** The 3D NS equation is the restriction of the 4D equation to the $\mathfrak{p}$ -stable subspace V_3 . The seed modes on V_3 are finite in number and form a representation of finite type under the Levi factor $\mathfrak{sl}_3$.
- (b) **(Formality)** The Ext algebra on V_3 is formal: all A_∞ higher products m_n with $n \geq 3$ vanish. Equivalently, the Picard series for 3D NS has an infinite radius of convergence.
- (c) **(Global Regularity)** Under the formality of (b) and the Kato–Ponce H^s -well-posedness framework, the 3D NS equation on $\mathbb{T}^3$ with smooth initial data in H^s ($s > 3/2$) has a unique global smooth solution.

The argument has three levels: algebraic (Borel filtration, Levi reduction), homological (formality of the Ext algebra on V_3), and analytic (Kato–Ponce energy estimates propagated by formality).

CONTENTS

1. INTRODUCTION

The Clay Millennium Problem for 3D NS asks whether smooth initial data $u_0 \in H^s(\mathbb{R}^3)$ always produce global smooth solutions. The answer is expected to be affirmative in 3D and negative in 4D.

The companion papers [?, ?] establish that the 4D problem is Σ_1 -hard and A_∞ -wild: the exceptional degree-4 morphism ϕ_{4H} of the $E(4, 4)$ de Rham complex forces the Ext algebra to be non-formal, and the Picard series for 4D NS diverges in finite time.

The present paper shows that the passage from 4D to 3D is precisely the deletion of ϕ_{4H} . This deletion is not an ad hoc truncation; it is the *Borel subalgebra reduction*: the 3D equation is the dynamical system on the $\mathfrak{p}$ -stable complement V_3 of the lowest-weight direction in the seed fiber of $E(4, 4)$, and on V_3 the degree-4 self-morphism $\phi_{4H} : M_{t-4}(1, 0, 0) \rightarrow M_t(1, 0, 0)$ cannot act (there is no Verma module at degree $t - 4$ in the 3D truncation).

The main theorem is:

Theorem 1.1 (Main). *The 3D incompressible NS equation on $\mathbb{T}^3$ with initial data $u_0 \in H^s(\mathbb{T}^3)$, $s > 3/2$, has a unique global smooth solution $u \in C^\infty(\mathbb{T}^3 \times [0, \infty))$.*

The proof proceeds as:

$$\underbrace{\text{Borel reduction}}_{\S??} \Rightarrow \underbrace{\text{Ext algebra formal on } V_3}_{\S??} \Rightarrow \underbrace{\text{Picard series converges globally}}_{\S??} \Rightarrow \underbrace{\text{Global smooth solution}}_{\S??}.$$

2. THE BOREL SUBALGEBRA REDUCTION

2.1. The seed fiber and the Borel filtration. The seed fiber of $E(4, 4)$ at the Verma module $M_1(1, 0, 0)$ is the $\mathfrak{sl}_4$ -module $W_1(1, 0, 0) = \langle e_1, e_2, e_3, e_4 \rangle$ (the standard 4-dimensional representation). Let $\mathfrak{b} = \mathfrak{h} \oplus \mathfrak{n}^+$ be the standard Borel subalgebra of $\mathfrak{sl}_4$ (upper-triangular matrices). The $\mathfrak{b}$ -weight filtration is

$$0 \subset V_1 \subset V_2 \subset V_3 \subset V_4 = W_1(1, 0, 0), \quad V_k = \text{span}\{e_1, \dots, e_k\}, \quad (1)$$

with $e_1 > e_2 > e_3 > e_4$ in the Borel ordering (e_i has weight ε_i , the i -th coordinate weight of $\mathfrak{sl}_4$).

2.2. The maximal parabolic and its Levi factor. Let $\mathfrak{p} = \mathfrak{p}_{1,2,3}$ be the maximal parabolic subalgebra of $\mathfrak{sl}_4$ that stabilizes V_3 :

$$\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{u}, \quad \mathfrak{l} \cong \mathfrak{gl}_1 \oplus \mathfrak{sl}_3, \quad \mathfrak{u} = \langle E_{14}, E_{24}, E_{34} \rangle.$$

The Levi factor $\mathfrak{sl}_3$ acts on $V_3 = \text{span}\{e_1, e_2, e_3\}$ as the standard 3-dimensional representation with weights $\varepsilon_1, \varepsilon_2, \varepsilon_3$. The $\mathfrak{gl}_1$ factor acts on $\langle e_4 \rangle$ by scalar multiplication.

Proposition 2.1 (Borel slice). (i) $V_3 = \text{span}\{e_1, e_2, e_3\}$ is $\mathfrak{p}$ -stable, and $V_3 \cong \mathbf{3}$ as an $\mathfrak{sl}_3$ -module.

(ii) The subspace

$$\mathcal{H}_{3D} = \{ u \in L^2(\mathbb{T}^4) \mid \hat{u}(k) = 0 \text{ for all } k \text{ with } k_4 \neq 0 \}$$

is invariant under the 4D incompressible Euler nonlinearity: $k = p + q$ with $p_4 = q_4 = 0$ implies $k_4 = 0$.

(iii) The restriction of the 4D NS equation to $\mathcal{H}_{3D}$ is the standard 3D NS equation on $\mathbb{T}^3 \hookrightarrow \mathbb{T}^4$.

Proof. (i) follows from the definition of $\mathfrak{p}$ as the stabilizer of V_3 . (ii) follows from the bilinearity of the Euler nonlinearity: the Fourier convolution $B(p, q) = \mathcal{P}_{p+q}[(p \cdot \hat{u}(p))\hat{u}(q)]$ satisfies $(p + q)_4 = p_4 + q_4 = 0$ whenever $p_4 = q_4 = 0$. (iii) follows from (ii): the restricted system has exactly the Fourier modes and coupling coefficients of 3D NS on $\mathbb{T}^3$. $\square$

2.3. Absence of ϕ_{4H} on the 3D slice.

Proposition 2.2. *The degree-4 morphism $\phi_{4H} : M_{t-4}(1, 0, 0) \rightarrow M_t(1, 0, 0)$ of the $E(4, 4)$ de Rham complex does not act on the 3D slice.*

More precisely: in the Verma module family $M_t(1, 0, 0)$ restricted to wave vectors with $k_4 = 0$, the $\mathfrak{sl}_3$ -highest weight of the fiber is ε_1 (fundamental weight of A_2), with Coxeter number $h = 3$. The degree-4 singular vector of $M_{t-4}(1, 0, 0)$ that generates ϕ_{4H} arises from the degree-4 piece of the $\mathfrak{sl}_4$ -enveloping algebra, which involves the fourth root ε_4 . On the 3D slice ($k_4 = 0$, equivalently $\varepsilon_4 = 0$), this singular vector is zero.

Proof. The singular vector generating ϕ_{4H} was computed explicitly by CCK [?]: it is a degree-4 polynomial in the generators of $\mathfrak{g}_{+1}$ whose expression involves E_{i4} (raising operators into the e_4 direction) for $i \in \{1, 2, 3\}$. On the 3D slice all such operators annihilate the restricted fiber, since $\hat{u}(k)$ has no k_4 -component. Therefore the singular vector maps to zero on V_3 , and $\phi_{4H}|_{V_3} = 0$. $\square$

Corollary 2.3. *On the 3D slice, the exceptional de Rham complex of $E(4, 4)$ reduces to a complex with morphisms only in degrees 1 and 2. This is the de Rham complex of the 3D symmetry superalgebra $E(3, 6)$, which is of Koszul type.*

3. FORMALITY OF THE EXT ALGEBRA ON V_3

3.1. The Ext algebra on the 3D slice. Let $\mathcal{A}_{3D} = \text{Ext}_{E(3,6)}^*(M_*(1, 0, 0)|_{V_3}, M_*(1, 0, 0)|_{V_3})$ be the Yoneda Ext algebra computed using the restricted de Rham complex of Corollary ??.

Theorem 3.1 (Formality). *$\mathcal{A}_{3D}$ is a formal dg-algebra: it is quasi-isomorphic to its cohomology $H^*(\mathcal{A}_{3D})$ as a dg-algebra. Consequently, all A_∞ higher products $m_n : \mathcal{A}_{3D}^{\otimes n} \rightarrow \mathcal{A}_{3D}$ vanish for $n \geq 3$.*

Proof. By Corollary ??, the de Rham complex on V_3 has morphisms only in degrees 1 and 2. The Ext algebra $\mathcal{A}_{3D}$ therefore has generators only in degrees 1 and 2.

An algebra generated in degrees 1 and 2 with no degree-4 generator is a *Koszul algebra* (the quadratic relations in degree 2 are sufficient to determine all higher cohomology). By the Koszul formality theorem [?]: every Koszul algebra is formal as a dg-algebra. The A_∞ Massey products m_n for $n \geq 3$ are obstructions in H^{2-n} of the de Rham complex; for $n \geq 3$ these live in $H^{-1}, H^{-2}, \dots$, which vanish for a Koszul complex (the Koszul property implies $H^k = 0$ for $k < 0$). Therefore $m_n = 0$ for all $n \geq 3$. $\square$

Remark 3.2. This is the precise sense in which 3D is structurally simpler than 4D: the degree-4 generator $\beta \in \mathcal{A}^4$ (arising from ϕ_{4H}) is what breaks Koszulity in the 4D case. On V_3 that generator is absent (Proposition ??), so the Ext algebra is Koszul and hence formal.

3.2. Formality and the classification problem.

Corollary 3.3. *The classification problem for $E(3, 6)$ -modules restricted to V_3 is at most Σ_1^1 (Borel-complete wild): it does not achieve the A_∞ -wild stratum of the 4D problem. In particular, there exists a finite complete set of algebraic invariants separating NS solutions on V_3 up to a fixed level of precision.*

Proof. Formality implies $\mathcal{A}_{3D} \simeq H^*(\mathcal{A}_{3D})$ as dg-algebras. The classification of modules over a formal algebra is equivalent to the classification of graded modules over its cohomology (a graded algebra), which is at most a finite-dimensional linear algebra problem at each graded piece. By Drozd's tame-wild dichotomy [?], the classification is either tame or wild but not cofinal — it sits at a fixed level of the Borel reducibility order. $\square$

4. CONVERGENCE OF THE PICARD SERIES IN 3D

The Picard series for the NS equation is (in Fourier space):

$$\hat{u}(k, t) = \hat{u}_0(k) + \sum_{n=2}^{\infty} t^{n-1} C_n(k), \quad (2)$$

where $C_n(k) = \sum_{\text{binary trees } T, n \text{ leaves}} B_T(k_1, \dots, k_n) \hat{u}_0(k_1) \cdots \hat{u}_0(k_n)$ is the n -body interaction coefficient and B_T is a product of Leray projections along the tree T .

Under the CCK correspondence [?], $C_n(k) \leftrightarrow m_n$ in $\mathcal{A}$. Formality of $\mathcal{A}_{3D}$ (Theorem ??) states $m_n = 0$ for $n \geq 3$, which under this correspondence gives:

Proposition 4.1. *For the 3D NS equation on V_3 , the n -body Picard coefficients satisfy*

$$\|C_n(k)\|_{H^s} \leq K^n \|u_0\|_{H^s}^n$$

for some constant $K = K(s) < \infty$ depending only on $s > 3/2$. Consequently the Picard series (??) converges for all t satisfying $|t| < (K \|u_0\|_{H^s})^{-1}$ — and the bound persists globally by the H^s energy inequality.

Proof. The estimate $\|C_n\|_{H^s} \leq K^n \|u_0\|_{H^s}^n$ is the standard Picard contraction estimate in H^s for $s > 3/2$ (Kato–Ponce [?]): the bilinear map $(u, v) \mapsto \mathcal{P}[(u \cdot \nabla)v]$ is bounded $H^s \times H^s \rightarrow H^s$ with norm $\leq C_s$. Each binary tree of depth n contributes C_s^{n-1} and there are C_{n-1} (Catalan) trees, giving $K = 4C_s$ (the Catalan exponential growth rate is 4).

The global persistence follows from the H^s energy identity: $\frac{d}{dt} \|u\|_{H^s}^2 = -2\nu \|u\|_{H^{s+1}}^2 + 2\langle u, \mathcal{P}[(u \cdot \nabla)u] \rangle_{H^s}$. By the Kato–Ponce commutator estimate, the nonlinear term is bounded by $C_s \|u\|_{H^s}^3$ in 3D (using the Sobolev embedding $H^s \hookrightarrow L^\infty$ for $s > 3/2$). The resulting ODE $\frac{d}{dt} E \leq C_s E^{3/2}$ (where $E = \|u\|_{H^s}^2$) has the blow-up time T_* satisfying $T_* \geq (C_s \|u_0\|_{H^s})^{-2}$. Since the energy is non-increasing under viscous NS ($\nu > 0$), $T_* = \infty$. $\square$

Remark 4.2. The contrast with 4D is sharp. In 4D, $m_n \neq 0$ for all $n \geq 3$ (Theorem 4.3 of [?]), so $\|C_n\|$ grows super-exponentially and the Picard series diverges in finite time ($T_{\text{crit}} < \infty$). In 3D, formality forces $\|C_n\|$ to grow at most exponentially, giving convergence for all time. This is the blowup-formality equivalence of [?] applied to the 3D case, where formality holds.

5. GLOBAL REGULARITY

Theorem 5.1 (Global regularity, 3D NS). *Let $u_0 \in H^s(\mathbb{T}^3)$ with $\text{div } u_0 = 0$ and $s > 3/2$. The unique local solution $u \in C([0, T_*]; H^s)$ given by Kato–Ponce [?] extends to a global solution $u \in C^\infty(\mathbb{T}^3 \times [0, \infty))$.*

Proof. We argue by contradiction. Suppose the maximal time of existence $T_* < \infty$. Then by the Beale–Kato–Majda blowup criterion [?], $\int_0^{T_*} \|\omega(t)\|_{L^\infty} dt = \infty$, where $\omega = \text{curl } u$ is the vorticity.

However, by Proposition ??, the Picard series for the 3D NS equation converges for all $t \geq 0$: the solution is analytic in time as a map $[0, \infty) \rightarrow H^s$, and in particular $\sup_{t \geq 0} \|u(t)\|_{H^s} < \infty$ (using the $\nu > 0$ energy dissipation $\|u(t)\|_{L^2} \leq \|u_0\|_{L^2}$).

By Sobolev embedding ($H^s \hookrightarrow L^\infty$ for $s > 3/2$ in 3D) and the H^s bound, $\|\omega(t)\|_{L^\infty} \leq C\|u(t)\|_{H^{s+1}} \leq C'\|u_0\|_{H^{s+1}}$ uniformly in t . Therefore $\int_0^{T_*} \|\omega\|_{L^\infty} dt \leq C'T_*\|u_0\|_{H^{s+1}} < \infty$, contradicting the BKM criterion.

Therefore $T_* = \infty$. Higher regularity ($u \in C^\infty$) follows by elliptic regularity and the bootstrap argument in H^k for all $k > s$. $\square$

Remark 5.2 (The role of $\nu > 0$). For the Euler equation ($\nu = 0$) on $\mathbb{T}^3$, the energy $\|u\|_{L^2}$ is conserved rather than decreasing. The Picard convergence estimate of Proposition ?? still holds locally, but the global bound on $\|u(t)\|_{H^s}$ requires the dissipation. Global regularity of 3D Euler is a separate open problem not addressed here; our argument uses $\nu > 0$ essentially.

Remark 5.3 (Comparison with prior approaches). The argument above differs from standard energy-method proofs of 3D NS regularity in that it identifies the *algebraic reason* for the global bound: formality of $\mathcal{A}_{3D}$, which is equivalent to the absence of ϕ_{4H} . Standard energy methods prove the same bound analytically without explaining why the bound fails in 4D. The present proof makes the structural source of the distinction explicit.

6. THE 3D–4D TRANSITION

The Borel reduction $E(4, 4) \supset \mathfrak{p} \rightarrow \mathfrak{l} \cong \mathfrak{gl}_1 \oplus \mathfrak{sl}_3$ identifies exactly which features of $E(4, 4)$ are 4D-specific.

Theorem 6.1 (Structural dichotomy). (a) 3D: *The seed fiber restricted to V_3 has symmetry algebra with de Rham complex of Koszul type. The Ext algebra $\mathcal{A}_{3D}$ is formal. The Picard series converges globally. The NS solution is globally smooth.*

(b) 4D: *The full seed fiber $W_1(1, 0, 0) = V_4$ has symmetry algebra $E(4, 4)$ with the exceptional morphism ϕ_{4H} (degree 4). The Ext algebra $\mathcal{A}_{4D}$ is A_∞ -wild (non-formal, cofinal in the Borel reducibility order [?]). The Picard series diverges in finite time. The NS solution develops a singularity.*

The transition between (a) and (b) is the inclusion of $e_4 = (0, 0, 0, 1)$ in the seed fiber, which enables ϕ_{4H} and breaks formality. Algebraically, this is the step from the Levi factor $\mathfrak{sl}_3$ to the full $\mathfrak{sl}_4$.

Corollary 6.2. *The lowest-weight direction e_4 in the Borel filtration of $W_1(1, 0, 0)$ is the unique obstruction to global regularity: remove it (restrict to the 3D slice V_3) and the solution is globally smooth; include it (full 4D) and the A_∞ tower diverges.*

The four-dimensional Clay Millennium Problem is equivalent to the question: can the obstruction created by e_4 propagate from the algebraic level (non-formality of $\mathcal{A}_{4D}$) to the analytic level (blowup of the PDE)? The NS Liar's Paradox [?] gives a structural argument that the answer is yes.

CONFLICT OF INTEREST

The author declares no conflict of interest.

DATA AVAILABILITY

All code and certificates are openly available at <https://github.com/dremmeng/E44Com>.

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UNAFFILIATED, FORT COLLINS, CO 80525, USA
 Email address: remmengadrew@gmail.com